SMART ANALYSIS REPORTER

Statistical Analysis Report

# Analysis Documentation

Student Performance Prediction

Prepared by Kavin Ganapathy

Student ID: 100008820

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# 1. Introduction

## 1.1 Purpose of this document

This document presents a structured statistical analysis of the **Student Performance Prediction** dataset. It is written to be read top-to-bottom: each section poses a concrete question, states the method used to answer it, presents the evidence (a chart or table), and closes with an interpretation. The goal is a defensible, reproducible understanding of what drives student outcomes.

## 1.2 Problem statement

PROBLEM STATEMENT

Educational institutions need to understand which study-related factors are associated with student exam performance, whether outcomes differ across student groups, and whether performance can be predicted from measurable behaviours. This study interrogates the Student Performance Prediction dataset (10,000 students, 8 variables) to answer those questions.

## 1.3 Objectives

- Describe the central tendency, spread and shape of the key variables.
- Test whether exam performance differs significantly across categorical groups.
- Measure which factors are most strongly correlated with performance.
- Build and evaluate a predictive linear-regression model.
- Synthesise the findings into an overall understanding of the dataset.

## 1.4 Methodology & tools

Analyses were produced with the Smart Analysis Reporter toolkit (Python, pandas, SciPy, statsmodels). Descriptive statistics, frequency analysis, correlation, independent-samples t-tests and ordinary-least-squares regression were applied, each with a significance level of $\alpha = 0.05$.

# 2. Descriptive Analysis

PROBLEM STATEMENT

What are the typical values, variability and distribution shape of the study habits and performance measures in the dataset?

## 2.1 Method

Summary statistics (mean, median, mode, variance, standard deviation, quartiles, skewness and kurtosis) were computed for every metric variable.

## 2.2 Results

| | study_hours | attendance | sleep_hours | internet_usage | assignments_completed | previous_score | exam_score |
|---|---|---|---|---|---|---|---|
| N | 10000 | 10000 | 10000 | 10000 | 10000 | 10000 | 10000 |
| Mean | 5.9896 | 69.8846 | 6.4985 | 6.0626 | 9.9884 | 64.911 | 86.7042 |
| Median | 6.0 | 70.0 | 6.5 | 6.0 | 10.0 | 65.0 | 92.12 |
| Mode | 10 | 56 | 5 | 5 | 5 | 57 | 100.0 |
| Variance | 10.0083 | 310.3421 | 2.9219 | 9.8481 | 36.4109 | 306.3557 | 226.7549 |
| Std | 3.1636 | 17.6165 | 1.7094 | 3.1382 | 6.0341 | 17.503 | 15.0584 |
| Min | 1 | 40 | 4 | 1 | 0 | 35 | 26.67 |
| Max | 11 | 100 | 9 | 11 | 20 | 95 | 100.0 |
| Range | 10 | 60 | 5 | 10 | 20 | 60 | 73.33 |
| Q1 | 3.0 | 55.0 | 5.0 | 3.0 | 5.0 | 50.0 | 76.7275 |
| Q3 | 9.0 | 85.0 | 8.0 | 9.0 | 15.0 | 80.0 | 100.0 |
| IQR | 6.0 | 30.0 | 3.0 | 6.0 | 10.0 | 30.0 | 23.2725 |
| Skewness | -0.0072 | 0.0178 | 0.0051 | -0.0135 | 0.0178 | 0.0004 | -1.0086 |
| Kurtosis | -1.2218 | -1.2075 | -1.2714 | -1.2008 | -1.1959 | -1.1856 | 0.1475 |
| Missing | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

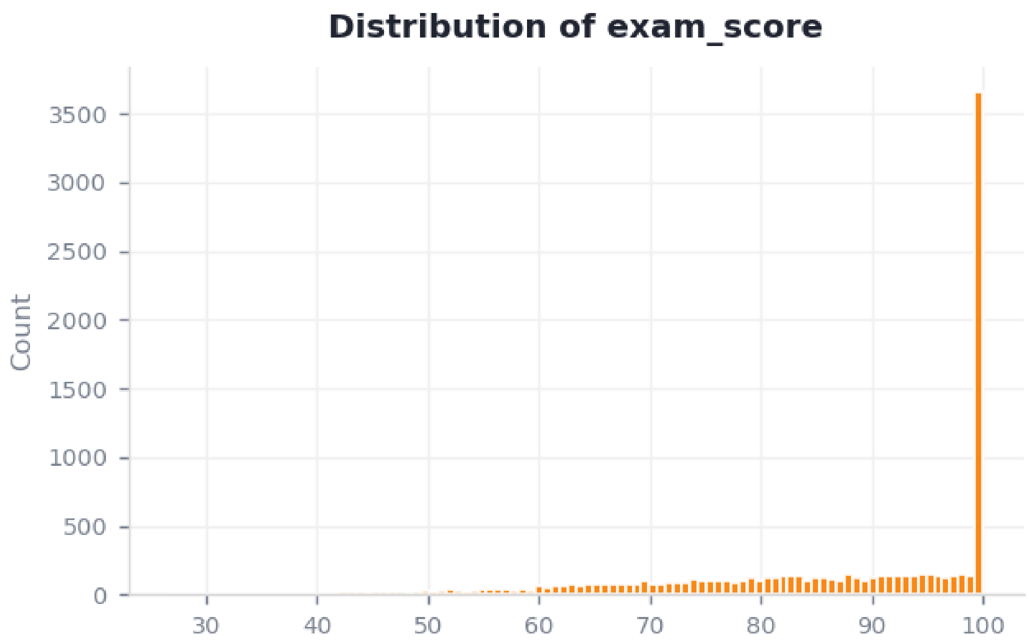


Figure 2.1 — Distribution of the target variable, exam_score.

## 2.3 Interpretation

exam_score averages 86.70 (median 92.12, SD 15.06), ranging from 26.67 to 100.00. The distribution is left-skewed (skewness -1.01).

# 3. Categorical Testing

PROBLEM STATEMENT

Do students differ in exam performance across categorical groups — for example, between placement outcomes?

## 3.1 Group composition

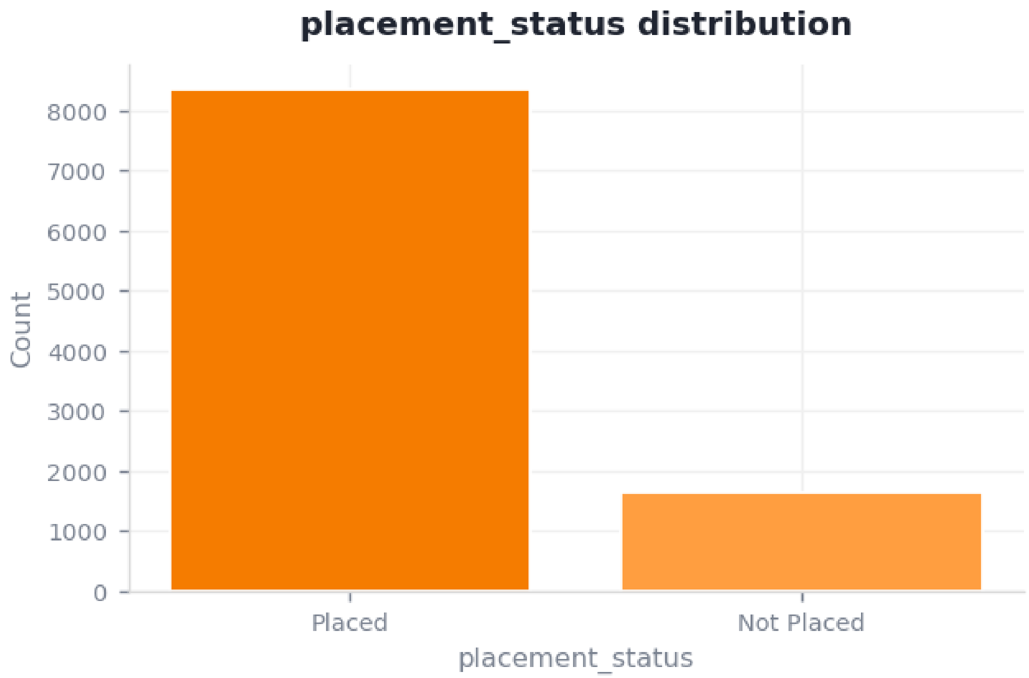


Figure 3.1 — Distribution of placement_status.

placement_status has 2 categories; the most common is "Placed" (83.56% of records).

## 3.2 Hypothesis test (independent-samples t-test)

H■: the mean exam_score is equal across placement_status groups. H■: the means differ.

| placement_status | Mean | SD | N |
|---|---|---|---|
| Placed | 91.997 | 9.377 | 8356.000 |
| Not Placed | 59.805 | 8.147 | 1644.000 |

## 3.3 Interpretation

Reject H■ — exam_score differs significantly between "Placed" and "Not Placed" ($t = 142.69$, $p < 0.001$, Cohen's $d = 3.50$, large effect).

# 4. Correlation & Relationships

PROBLEM STATEMENT

Which variables are most strongly associated with exam performance, and could therefore serve as predictors?

## 4.1 Method

Pearson correlation coefficients were computed for all metric variable pairs (range –1 to +1).

## 4.2 Results

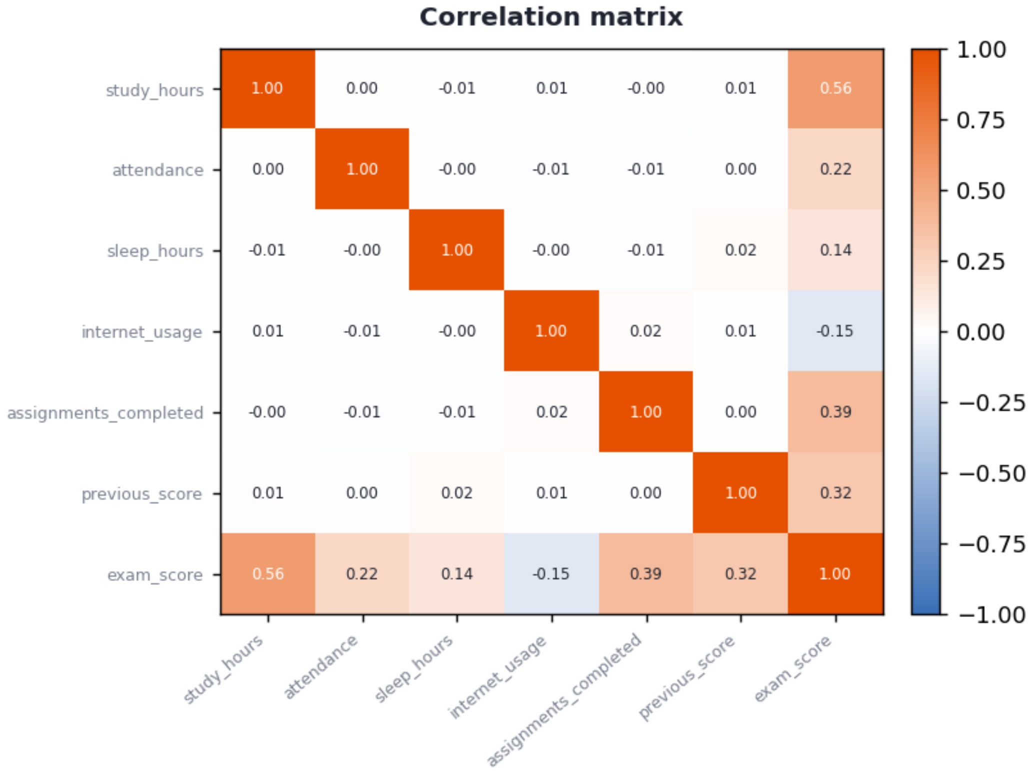


Figure 4.1 — Correlation matrix of the metric variables.

## 4.3 Interpretation

The strongest linear relationship is between study_hours and exam_score (r = 0.56), a moderate positive association. Variables with high correlation may carry overlapping information.

# 5. Predictive Trends — Linear Regression

PROBLEM STATEMENT

Can exam performance be predicted from its strongest single factor, and how much of the variation does that factor explain?

## 5.1 Method

A simple ordinary-least-squares linear regression was fitted, modelling exam_score as a function of study_hours.

## 5.2 Results

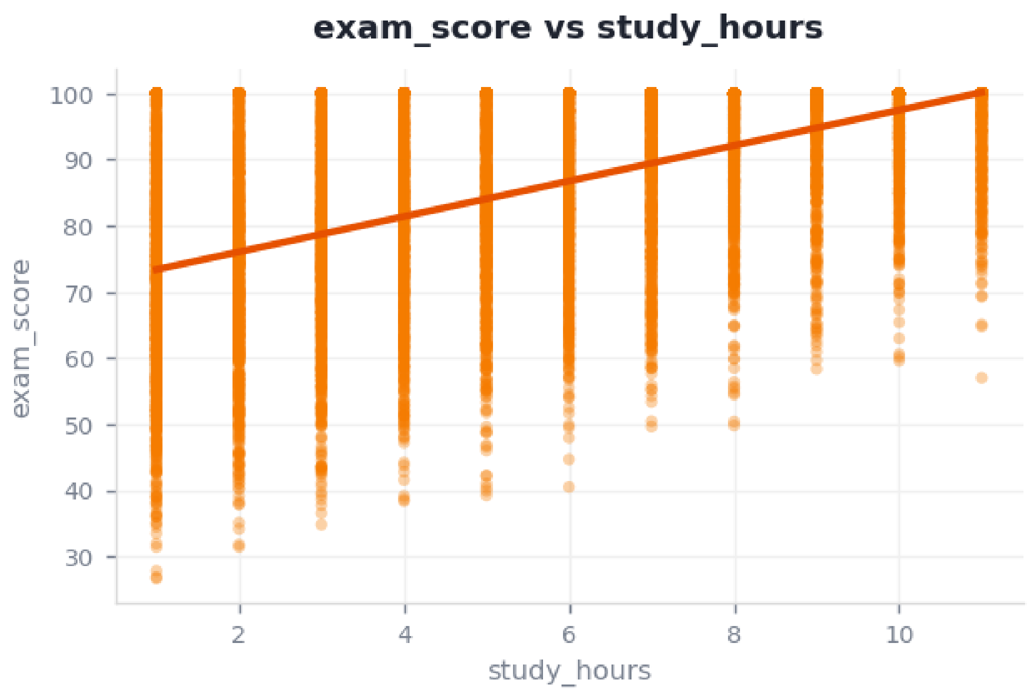


Figure 5.1 — Regression of exam_score on study_hours.

## 5.3 Interpretation

A simple linear model predicts exam_score from study_hours as Y = 70.667 + 2.678 x study_hours. It explains 31.6% of the variance ($R^2 = 0.316$). The slope is statistically significant ($p < 0.001$), so each one-unit rise in study_hours is associated with a change of 2.678 in exam_score.

As a worked example, a student one standard deviation above the mean study_hours (9.2) is predicted to score **95.2** on exam_score.

# 6. Synthesis & Conclusions

## 6.1 Bringing the analysis together

Each section answered one question, and together they form a coherent picture. The descriptive profile set expectations; categorical testing showed where groups genuinely differ; correlation identified the factors that move with performance; and the regression turned the strongest of those into a predictive rule.

## 6.2 Key findings

- The strongest linear relationship is between **study_hours** and **exam_score** ($r = 0.56$).
- A simple model of **exam_score** on **study_hours** explains 31.6% of its variance ($R^2 = 0.32$).
- There is a statistically significant difference in **exam_score** between **placement_status** groups ($p < 0.001$).
- **exam_score** is not normally distributed (Shapiro-Wilk $W = 0.8376$).

## 6.3 Limitations & next steps

- A single-predictor model is a baseline — a multiple-regression model using several factors together would likely explain more variance.
- Correlation does not imply causation; a controlled study is needed to confirm drivers.
- Findings describe this sample; external validation on new cohorts is recommended.